QUANTIFYING SHRIMP WELFARE THREATS

THE BIGGEST SOURCES OF SHRIMP SUFFERING

Comparing **18 welfare threats** to understand where the greatest welfare harms may occur.



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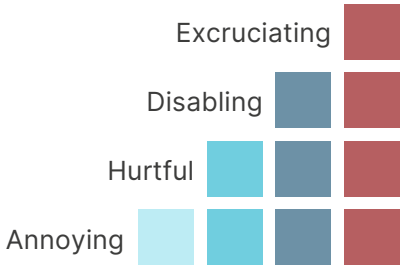
HOW DO YOU COMPARE SUFFERING?

Rethink Priorities models each welfare threat across three dimensions.

1 **HOW MANY**
SHRIPM ARE AFFECTED?

2 **HOW LONG**
DOES THE HARM LAST?

3 **HOW SEVERE**
IS THE PAIN?



Pain across severity levels is converted into a common metric: **disabling-equivalent pain.**

Uncertainty is built in. The model uses ranges rather than treating every estimate as certain.

18 THREATS. DIFFERENT FORMS OF HARM.

From water quality and overcrowding to transport, hunger and slaughter, shrimp can face harm in very different ways.

WATER QUALITY

- Low dissolved oxygen
- High temperature
- Low temperature
- Low salinity
- Nonoptimal pH
- High un-ionized ammonia
- Pollutants

HEALTH

- Eyestalk ablation
- Biosecurity failures

TRANSPORT

- Waterless transport
- Water-based transport

POND HABITAT

- High density
- Lack of substrate
- Predators

END OF PRODUCTION

- Harvest
- Slaughter

FEED MANAGEMENT

- Malnutrition
- Hunger



HIGHER ESTIMATED IMPACT — Chronic + Widespread

LOWER ESTIMATED IMPACT — Acute + Limited

THE BIGGEST HARM MAY HAPPEN SLOWLY.

The highest mean estimates come from conditions that affect many shrimp over long periods.

64 h
HIGH STOCKING DENSITY
5.2 — 112

30 h
HIGH UN-IONIZED AMMONIA
0.75 — 108

18 h
LOW DISSOLVED OXYGEN
1.8 — 57

157 h
ESTIMATED DISABLING-EQUIVALENT PAIN PER AVERAGE FARMED SHRIMP IN THE MODEL
28 — 356

Widespread, **long-lasting harms** can add up to greater overall suffering.

All hours refer to estimated disabling-equivalent pain.
● Mean estimate · Line = 90% subjective credible interval.

THE SCALE CHANGES THE PICTURE.

Individual harms can seem small or invisible. Across hundreds of billions of shrimp, they add up.

95

TRILLION HOURS / YEAR

17T — 216T

90% SCI

Assuming **~608 billion penaeid shrimp die on ongrowing farms annually**, mean estimates imply 95 trillion hours of disabling-equivalent pain per year.



1 WHAT THE MODEL SUGGESTS

Chronic, widespread threats may account for particularly large amounts of suffering.

2 WHAT REMAINS UNCERTAIN

The evidence does not allow most welfare threats to be ranked confidently.

Better evidence can help **bring hidden harms into focus.**

ABOUT THE RESEARCH

Based on *“Quantifying and prioritizing shrimp welfare threats”*
by Hannah McKay and William McAuliffe,
Rethink Priorities, 2024.

READ THE FULL RESEARCH →

This visual summary highlights selected findings and does not replace the full report. The model focuses on penaeid shrimp in ongrowing farms and broodstock facilities and does not cover all possible welfare threats or earlier production stages. Estimates involve substantial uncertainty; see the original research for methodology, assumptions, and limitations.

Visual summary & design by María Vera.

Independent concept project. Not commissioned by Rethink Priorities.

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